Statistical Arbitrage in US Equities

Avellaneda & Lee [2010]

STATARB RESEARCH

IEOR 4733 - ALGORITHMIC TRADING

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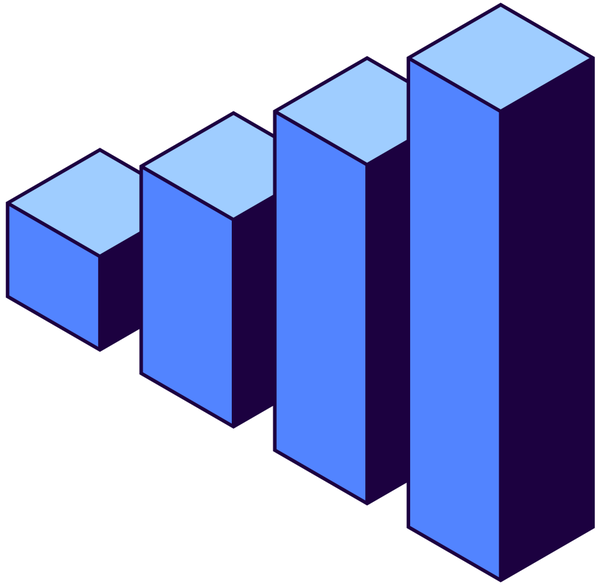


Proposal

EXTENSIONS

HMM Regime Filter
Regime Detection

Almgren–Chriss Execution
Optimal Execution

Vol-Targeted Sizing
Risk Management

Topic

- Reproduction and extension of the Avellaneda & Lee (2010) statistical arbitrage framework for market-neutral equity trading using PCA-based factor decomposition and mean-reverting residual models.

Introduction

- We reproduce the Avellaneda-Lee framework for market-neutral statistical arbitrage. The core idea is to decompose stock returns into systematic factors and idiosyncratic residuals, model the residuals as mean-reverting Ornstein-Uhlenbeck processes, and trade stocks whose residuals deviate significantly from equilibrium.
- We then extend this framework with three enhancements: a Hidden Markov Model for regime detection, the Almgren-Chriss optimal execution model for realistic transaction costs, and volatility-targeted position sizing for adaptive risk management.

Literature & Methods

Overview of Literature and the Methods

Literature:

Avellaneda & Lee (2010) built on mean-reversion research by Lo & MacKinlay (1990) and Poterba & Summers (1988). They showed PCA-extracted eigenportfolios serve as effective risk factors. Khandani & Lo (2007) documented the August 2007 quant crisis using related contrarian strategies. Our extensions draw on Almgren & Chriss (2001) for execution modeling and Hamilton (1989) for HMM regime switching.

Core Methodology:

1. Multi-factor return decomposition:

$$R_i = \sum_{j=1}^{m} \beta_{ij} F_j + \tilde{R}_i$$

2. Model residuals as Ornstein-Uhlenbeck process:

$$dX_i(t) = \kappa_i(m_i - X_i(t))dt + \sigma_i dW_i(t), \quad \kappa_i > 0$$

3. Compute s-scores and trade on thresholds:

$$s_i = \frac{X_i(t) - m_i}{\sigma_{eq,i}}, \quad \sigma_{eq,i} = \frac{\sigma_i}{\sqrt{2\kappa_i}}$$

Signal & Execution:

4. Trading rules: Enter trades when |s| > 1.25. Exit long at s > -0.50. Exit short at s < 0.75. Asymmetric exit from the paper's calibration.

5. Volume-adjusted returns: Rescale daily returns by inverse of recent trading volume to discount moves on heavy volume.

6. Market neutrality: Beta-hedged via SPY or sector ETFs. 2+2 leverage (2x long, 2x short).

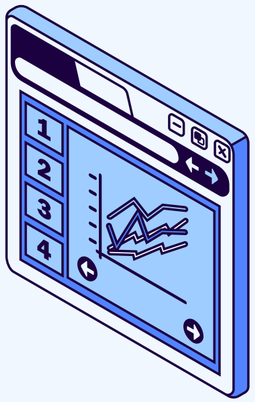
7. Estimation: 252-day window for correlation matrix. 60-day window for OU params. Stocks with kappa > 8.4 only.

Volume-adjusted returns: $\tilde{R}_t = R_t \times \left[\frac{\langle \delta V \rangle}{V(t+\Delta t) - V(t)}\right]$

Eigenportfolio returns: $F_{jk} = \sum_{i=1}^{N} \frac{v_i^{(j)}}{\sigma_i} R_{ik}$

Strategy Overview

Data, Methodology, and Evalution Metrics



Data	Methodology	Evaluation
Universe US equities, mkt cap > $1B at trade date (no survivorship bias)	**Factor Estimation** PCA on 252-day rolling correlation matrix; ETF single-stock regressions	**Sharpe Ratio** $(\mu - r) / \sigma$, per-strategy & per-sector
Price Data Adjusted close, dividends, corporate actions, split-adjusted	**Residual Modeling** OU process on 60-day window; s-scores with centred means	**Maximum Drawdown** Peak-to-trough; target < 15%
Volume Data Daily share volume; 10-day trailing avg for return rescaling	**Signal Generation** Open at $\lvert s \rvert > 1.25$, close at $\lvert s \rvert < 0.50$-0.75; filter by $\kappa > 8.4$	**Turnover & Costs** Chriss impact model; replaces flat 10bp
Factor Data 15 sector ETFs (XLF, XLK, XLE...) + risk-free rate	**Portfolio Construction** Market-neutral (beta ≈ 0); 2+2 leverage; bang-bang position sizing	**Out-of-Sample Design** Estimation uses prior 60 days only; zero lookahead

Strategy Overview (Cont.)

Weaknesses in the Current Method

- **Regime Blindness:** The OU model has no concept of market state. It kept generating mean-reversion signals straight through the August 2007 liquidation event, where the paper itself reports a ~10% drawdown across all strategy variants.

- **Stale Parameters:** A fixed 60-day estimation window can't adapt when the mean-reversion speed shifts abruptly. If kappa drops mid-window due to a structural break, the model is still trading on stale estimates for weeks.

- **Alpha Decay:** PCA-based strategies went from Sharpe ratios above 2.0 (2000-2002) to around 0.9 (2003-2007). The paper documents this decline but treats it as an observation rather than a problem to solve.

- **Static Costs:** A flat 10 bps round-trip treats a trade in AAPL the same as one in a $1B-cap regional bank. In practice, spread and impact costs vary by an order of magnitude across the universe.

- **Uniform Sizing:** A stock with 5% residual volatility gets the same dollar allocation as one with 15%. The result is that a handful of high-vol positions drive most of the portfolio's daily PnL swings.

Proposed Extensions

Extension 1: Regime Detection (HMM)

Fit a 2-state Hidden Markov Model on rolling residual volatility and cross-sectional dispersion features. Use smoothed state probabilities to halt or reduce entries during trending/crisis regimes.

$$P(z_t = \text{mean-rev} \mid y_{1:T}) \longrightarrow \text{Trade if } P > \theta$$

Extension 2: Execution Model (Almgren-Chriss)

Replace the flat 10 bps assumption with a model incorporating bid-ask spread and temporary price impact as a function of trade size and liquidity.

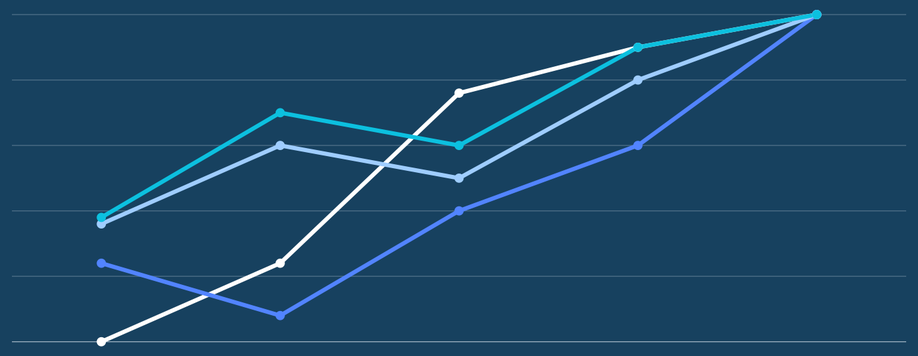
$$C(x) = \frac{1}{2}\gamma\sigma^2 \sum_k n_k^2 + \sum_k \eta \cdot |n_k| + \epsilon \sum_k |n_k|$$

Extension 3: Volatility-Targeted Position Sizing

Scale dollar exposure per trade inversely with recent realized volatility of the residual spread, keeping risk contribution per position roughly constant.

$$Q_i = \frac{\sigma_{\text{target}}}{\hat{\sigma} residual, i} \cdot Q\text{base}$$

Timeline



WEEK 1-2

Data pipeline and core reproduction. PCA implementation, OU estimation, s-score calculation. Validate intermediate outputs against the paper's reported eigenvalue spectrum.

WEEK 3

Backtest engine with proper rolling windows, transaction cost accounting, and portfolio tracking. Reproduce baseline ETF and PCA Sharpe ratios.

WEEK 4-5

Implement extensions: HMM regime filter, Almgren-Chriss execution model, vol-targeted sizing. Measure the incremental impact of each component.

WEEK 6

Build a Streamlit performance dashboard. Run sensitivity analysis on key parameters (number of PCA factors, s-score thresholds, estimation windows, cost assumptions).

WEEK 7

Final integration and testing. Deliver a fully functioning trading system pipeline that can run new simulations on fresh data. Prepare final presentation.

Technology Stack

- Python Libraries: NumPy, Pandas, Scikit-learn, HMMlearn, SciPy
- Dashboard: Streamlit
- Data: Yahoo Finance
- Visualization: Matplotlib/Plotly
- Version Control: Git

Thank You